

L9_S1_A.pdf
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Create a formal academic lecture scribe based strictly on the attached lecture slides and relevant textbook material. The scribe must serve as reliable reference material for a closed-notes examination, therefore it must prioritize correctness, precision, and strict faithfulness to the provided source material rather than creative explanation.

Content Requirements:

1. Include all definitions, symbols, and mathematical notations introduced in the lecture.
2. Clearly state all assumptions, propositions, theorems, or formal results exactly as presented in the lecture material.
3. Reconstruct all proofs, derivations, and worked examples step by step whenever they appear in the provided context.
4. Do not omit intermediate steps that are necessary to understand the reasoning shown in the lecture.

Restrictions:

1. Do not introduce new examples, explanations, or interpretations that are not present in the provided material.
2. Do not simplify concepts or provide intuitive explanations beyond what is explicitly given in the lecture.
3. If a concept, example, or derivation is not included in the context, do not generate it.

Structure:

1. Organize the scribe following the same logical sequence as the lecture.
2. Use clear section headings and subheadings to maintain the conceptual flow of the lecture.
3. Ensure the final document can be used directly by a student for accurate revision of the lecture content.



Below is a **formal lecture scribe constructed strictly from the provided lecture slides**. The structure, notation, derivations, and results follow the same order as the lecture material. No additional explanations or examples beyond the slides are introduced.

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace, and Gamma Random Variables

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Source: Lecture Slides  L9_S1_A

1. Types of Continuous Random Variables

The lecture introduces several types of continuous random variables and their probability distributions:

1. Uniform Random Variable
2. Exponential Random Variable
3. Laplace Random Variable
4. Gamma Random Variable

Additionally, examples and applications are provided for each distribution.  L9_S1_A

2. Uniform Random Variable

Definition

A random variable X is uniformly distributed over the interval $[a, b]$ if its probability density function (PDF) is

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x < b \\ 0, & \text{elsewhere} \end{cases}$$

The cumulative distribution function (CDF) is

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$$

The corresponding PDF and CDF graphs are illustrated in the lecture slides (Figure 3.8).

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3. Example 1 – Uniform Random Variable

Problem

The phase of a sinusoid θ is uniformly distributed over $[0, 2\pi]$.

Its PDF is

$$f_{\Theta}(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \leq \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

Find:

1. $Pr(\theta > 3\pi/4)$
2. $Pr(\theta < \pi \mid \theta > 3\pi/4)$
3. $Pr(\cos(\theta) < 1/2)$

Solution

(a) $Pr(\theta > 3\pi/4)$

For a uniform random variable on $[0, 2\pi]$,

$$Pr(a < \theta < b) = \frac{b-a}{2\pi}$$

Thus,

$$Pr(\theta > 3\pi/4) = \frac{2\pi - 3\pi/4}{2\pi} = \frac{5}{8}$$

(b) Conditional Probability

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

Let

$$A = (\theta < \pi)$$

$$B = (\theta > 3\pi/4)$$

$$Pr(3\pi/4 < \theta < \pi) = \frac{\pi - 3\pi/4}{2\pi} = \frac{1}{8}$$

$$Pr(\theta < \pi \mid \theta > 3\pi/4) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c) $Pr(\cos(\theta) < 1/2)$

$$\cos(\theta) = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Thus,

$$\cos(\theta) < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \theta < \frac{5\pi}{3}$$

Therefore,

$$Pr(\cos(\theta) < 1/2) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

Applications of Uniform Distribution

Examples include:

- Phase of a sinusoidal signal when all phases in $[0, 2\pi]$ are equally likely.
- Random number generation between 0 and 1 for simulations.
- Arrival time of a user within a known time window assuming no time preference.

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4. Exponential Random Variable

Definition

For $b > 0$, the exponential random variable has

PDF

$$f_X(x) = \frac{1}{b}e^{-x/b}u(x)$$

CDF

$$F_X(x) = [1 - e^{-x/b}]u(x)$$

where $u(x)$ is the unit step function.  L9_S1_A

5. Example 2 – Exponential Random Variable

Problem

Let X be an exponential random variable with

$$f_X(x) = e^{-x}u(x)$$

Find:

1. $Pr(3X < 5)$
 2. $Pr(3X < y)$
 3. Let $Y = 3X$. Find $f_Y(y)$.
-

Solution

Given

$$f_X(x) = e^{-x}u(x)$$

$$F_X(x) = [1 - e^{-x}]u(x)$$

(a)

$$Pr(3X < 5) = Pr(X < 5/3)$$

$$= F_X(5/3)$$

$$= 1 - e^{-5/3}$$

(b)

$$\begin{aligned}
 \Pr(3X < y) &= \Pr(X < y/3) \\
 &= F_X(y/3) \\
 &= [1 - e^{-y/3}]u(y)
 \end{aligned}$$

(c)

Let $Y = 3X$.From part (b), the CDF of Y is

$$F_Y(y) = [1 - e^{-y/3}]u(y)$$

Differentiating,

$$\begin{aligned}
 f_Y(y) &= \frac{d}{dy}F_Y(y) \\
 f_Y(y) &= \frac{1}{3}e^{-y/3}u(y)
 \end{aligned}$$

Thus Y is exponential with parameter

$$\lambda = \frac{1}{3}$$

Applications of Exponential Distribution

- Time until radioactive decay of a particle.
 - Duration of an idle period in a wireless communication channel.
 - Time until the next earthquake in simplified seismic models.  L9_S1_A
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6. Laplace Random Variable

Definition

PDF

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

CDF

$$F_X(x) = \begin{cases} \frac{1}{2} \exp(x/b), & x < 0 \\ 1 - \frac{1}{2} \exp(-x/b), & x \geq 0 \end{cases}$$

Application

The Laplace random variable is used to model the **amplitude of speech signals**.

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7. Example 3 – Laplace Random Variable

Problem

Let

$$f_W(w) = ce^{-2|w|}$$

Find:

1. c
 2. $Pr(-1 < W < 2)$
 3. $Pr(W > 0 \mid -1 < W < 2)$
-

Solution

(a) Find c

$$\int_{-\infty}^{\infty} ce^{-2|w|} dw = 1$$

$$2c \int_0^{\infty} e^{-2w} dw = 1$$

$$2c \left(\frac{1}{2} \right) = 1$$

$$c = 1$$

(b)

$$\begin{aligned}
 Pr(-1 < W < 2) &= \int_{-1}^0 e^{2w} dw + \int_0^2 e^{-2w} dw \\
 &= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4}) \\
 &= 1 - \frac{1}{2}(e^{-2} + e^{-4})
 \end{aligned}$$

(c)

$$\begin{aligned}
 Pr(W > 0 | -1 < W < 2) &= \frac{Pr(0 < W < 2)}{Pr(-1 < W < 2)} \\
 Pr(0 < W < 2) &= \int_0^2 e^{-2w} dw = \frac{1}{2}(1 - e^{-4})
 \end{aligned}$$

Thus,

$$Pr(W > 0 | -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

8. Gamma Random Variable

PDF

$$f_X(x) = \frac{(x/b)^{c-1} e^{-x/b}}{b\Gamma(c)} u(x)$$

CDF

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)} u(x)$$

Definitions

Gamma Function

$$\Gamma(a) = \int_0^\infty e^{-t} t^{a-1} dt$$

Incomplete Gamma Function

$$\gamma(a, \beta) = \int_0^\beta e^{-t} t^{a-1} dt$$

Special Cases

- $b = 2, c = 0.5 \rightarrow$ Chi-square random variable
- $c = 1 \rightarrow$ Exponential random variable

with

$$k = c, \quad \theta = b$$

9. Example 4 – Gamma Random Variable

Let

$$X \sim \text{Gamma}(\alpha, \lambda)$$

with PDF

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \geq 0$$

(a) Expected Value

$$\begin{aligned} E[X] &= \int_0^\infty x f_X(x) dx \\ E[X] &= \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1} dx \\ &= \frac{\Gamma(\alpha + 1)}{\lambda \Gamma(\alpha)} \\ E[X] &= \frac{\alpha}{\lambda} \end{aligned}$$

(b) Variance

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

First compute

$$E[X^2] = \frac{\Gamma(\alpha + 2)}{\lambda^2 \Gamma(\alpha)} = \frac{\alpha(\alpha + 1)}{\lambda^2}$$

Thus,

$$\text{Var}(X) = \frac{\alpha(\alpha + 1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2$$

$$\text{Var}(X) = \frac{\alpha}{\lambda^2}$$

10. Properties of the Gamma Function

Prove the following:

1.

$$\Gamma(n) = (n - 1)!$$

for $n = 1, 2, 3, \dots$

2.

$$\Gamma(x + 1) = x\Gamma(x)$$

3.

$$\Gamma(1/2) = \sqrt{\pi}$$

11. Applications of the Gamma Distribution

- Total service time for multiple independent tasks where each task duration is exponential.
- Received signal energy computed from multiple independent signal samples.
- Waiting time until the k -th event in a Poisson process.  L9_S1_A

12. Problem Solving Example – Movie Theater

Suppose arrival time depends on travel time T_r with PDF $f(t)$ and CDF $F(t)$.

Let d be the departure time.

Arrival time:

$$Arrival = d + T_r$$

Cost structure:

Early arrival:

$$cost = c(T - d - T_r)$$

Late arrival:

$$cost = k(d + T_r - T)$$

Expected cost:

$$E[C(d)] = \int_0^{T-d} c(T - d - t)f(t)dt + \int_{T-d}^{\infty} k(t + d - T)f(t)dt$$

Optimal Departure Time

Differentiate with respect to d and set equal to zero:

$$-cF(T - d) + k[1 - F(T - d)] = 0$$

Thus,

$$F(T - d^*) = \frac{k}{c + k}$$

$$d^* = T - F^{-1}\left(\frac{k}{c + k}\right)$$

Interpretation:

Depart such that the probability of arriving early equals

$$\frac{k}{c + k}$$

13. CDF Analysis Example

Given

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx}F_X(x) = 2x, \quad 0 \leq x \leq 1$$

Mean

$$\mu_X = \int_0^1 x(2x)dx$$

$$\mu_X = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)dx - \mu_X^2$$

$$= \frac{1}{2} - \frac{4}{9}$$

$$\sigma_X^2 = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X - \mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X - \mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

- Distribution is **left-skewed**
 - **Platykurtic** (lighter tails than Gaussian)
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If you want, I can also **convert this into LaTeX format ready for Overleaf submission**, which is usually what professors expect for **lecture scribe assignments in CSE courses**.

Sources



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